

Grad Café PostgreSQL Analysis Results

This report summarizes SQL analysis conducted on the Grad Café applicant dataset loaded into a PostgreSQL database. The database contains applicant records scraped and cleaned from Grad Café, including downloaded fields such as program, status, term, nationality, GPA, GRE scores, and degree type. It also includes LLM-generated fields for cleaned program and university names. The queries were written in `query_data.py` and the results were also displayed dynamically on the Flask webpage.

Required Analysis Questions

1. How many entries do you have in your database who have applied for Fall 2026?

Answer: 1702

Query description:

- I used `COUNT(*)` with a `WHERE` filtering for `term = 'Fall 2026'`.
- This counts only the applicant records where the start term was Fall 2026.

2. What percentage of entries are from international students?

Answer: 69.21%

Query description:

- I calculated the percentage of international entries using a `CASE` statement. While treating records marked as International as international entries, I also checked the `comments` field for nationality-related clues such as “Canadian” or “International,” because some self-submitted records include nationality information in the comments rather than in the structured applicant-type field.
- The denominator includes entries with usable nationality information, such as American, International, Other, or nationality clues in the comments.
- This approach was used because the scraped dataset contains some missing or unclear nationality values.

3. What is the average GPA, GRE, GRE V, and GRE AW of applicants who provide these metrics?

Answer:

Average GPA: 3.71

Average GRE Quant: 166.3

Average GRE Verbal: 158.76

Average GRE AW: 4.16

Query description:

- I used `AVG()` to calculate the average applicant metrics.
- For GPA, I used a normalized GPA expression so that values on 5.0 and 10.0 scales could be converted onto a 4.0 scale where possible.
- For GRE Quant and GRE Verbal, I only included values between 130 and 170, since those are the valid score ranges for each section. For GRE Analytical Writing, I included values between 0.1 and 6.0 and excluded zero values because they likely represent missing or unusable scraped data.

4. What is the average GPA of American students in Fall 2026?

Answer: 3.79

Query description:

- I filtered the database to Fall 2026 entries where the applicant was marked as American or where the comments included "US Citizen." I then calculated the average normalized GPA for those records.
- I used the normalized GPA expression for consistency with the overall GPA calculation.

5. What percent of entries for Fall 2026 are acceptances?

Answer: 44.83%

Query description:

- I filtered the table to records where term = 'Fall 2026'.
- Then I used a CASE statement to count records where status = 'Accepted', divided that count by the total number of Fall 2026 records, multiplied by 100, and rounded the result to two decimal places.

6. What is the average GPA of applicants who applied for Fall 2026 who are acceptances?

Answer: 3.74

Query description:

- I filtered to Fall 2026 records where status = 'Accepted', then calculated the average normalized GPA. This gives the average GPA among accepted Fall 2026 applicants who provided GPA values that could be interpreted and normalized.

7. How many entries are from applicants who applied to JHU for a master's degree in Computer Science?

Answer: 115

Query description:

- I counted records where the degree field matched a master's degree, the LLM-generated program matched Computer Science, and the LLM-generated university matched Johns Hopkins.
- I used the LLM-generated program and university fields because they separate the program and university names more cleanly than the original downloaded program field.

8. How many entries from 2026 are acceptances from applicants who applied to Georgetown University, MIT, Stanford University, or Carnegie Mellon University for a PhD in Computer Science?

Answer: 15

Query description:

- I filtered for 2026 records where the status was Accepted, the degree was PhD, and the downloaded program field contained Computer Science. Since the downloaded field combines program and university information, I used ILIKE filters to search for Georgetown University, MIT, Stanford University, Carnegie Mellon University, and CMU within the downloaded program text.

9. Do the numbers for Question 8 change if you use LLM-generated fields rather than downloaded fields?

Answer: 15. The number did not change.

Query description:

- I repeated Question 8's logic, but instead of searching the downloaded program field, I used the LLM-generated program and university fields.
- The result was still 15, which means that for this specific query, the cleaner LLM-generated fields did not change the count.
- Possible reasons for the lack of change:
 - o My original downloaded-field query already accounted for multiple versions of the university names. For example, I searched for both full names and common abbreviations such as "Massachusetts Institute of Technology" and "MIT," as well as "Carnegie Mellon" and "CMU." Because of this, the downloaded-field query may have already captured the same records that the LLM-generated university field later standardized.
 - o Another possible reason is that the final matching group was small: only 15 entries met all the criteria. With such a small result set, there may have been fewer messy or ambiguous university/program names for the LLM-generated fields to correct. Therefore, in this case, the LLM fields improved cleanliness and readability, but they did not change the final count.

Additional Analysis Questions

10. Which universities appear most competitive in Fall 2026 based on low acceptance rate?

I wanted to look beyond just counting acceptances and see which universities appeared more competitive in the Fall 2026 Grad Café data. I used acceptance rate instead of total acceptances because total counts alone can be misleading. I also only included universities with at least 10 entries so that the results were not overly affected by very tiny sample sizes. From the results, schools like Rensselaer Polytechnic Institute, University of Washington, Washington University in St. Louis, and Harvard appeared to have some of the lowest acceptance rates in this dataset. However, I would only interpret this as competitiveness within the Grad Café sample, not as official admissions rates.

University	Total Entries	Accepted	Acceptance Rate
Rensselaer Polytechnic Institute	12	1	8.33%
University of Washington	58	5	8.62%
Washington University In St. Louis (Washu/Wustl)	11	1	9.09%
Harvard University	24	3	12.50%
University of Colorado Boulder	24	4	16.67%
Yale University	28	5	17.86%
University of Pennsylvania	14	3	21.43%
University of Virginia	23	5	21.74%
Stanford University	67	15	22.39%
McGill University	13	3	23.08%

11. Which degree type had the highest Fall 2026 acceptance rate?

I wanted to compare acceptance rates across different degree types for Fall 2026. I chose this question because the dataset contains multiple degree categories, and I was curious whether master's, PhD, or other degree types had noticeably different acceptance patterns. I grouped Fall 2026 records by degree, counted total entries and accepted entries, and calculated the acceptance rate for each group. I only included degree types with at least 10 entries so that very small categories would not distort the comparison. The results show which Masters degree type had the highest acceptance rate within the Grad Café sample.

Degree Type	Total Entries	Accepted	Acceptance Rate
Masters	579	397	68.57%
PhD	1119	363	32.44%

Limitations

Some limitations that self-submitted datasets such as Grad Café contain include data not being a representation of the full applicant population because only some applicants choose to post their results. This creates a self-selection bias where we only see a biased group of applicants, mostly who are active in online admissions communities may be more likely to submit entries. The dataset also contains missing, inconsistent, and imperfectly formatted values, such as missing GPA or GRE scores, unclear nationality categories, nationality in the comments field and combined program-university strings. Because this data is anonymous and self-reported, there is no reliable way to verify whether every entry is accurate.

One result that stood out to me was how misleading the GPA analysis became before cleaning and normalization. When I first averaged the raw GPA field, the result was much lower than expected, which did not make sense for graduate admissions data. After checking the values more carefully, I realized that the GPA field was not consistently reported on one scale. Some applicants appeared to use a 4.0 scale, while others may have used 5.0 or 10.0 grading systems, and some values were missing or unclear.

Query

Query History

1

SELECT

2

COUNT(*) AS total_entries,

3

COUNT(gpa) AS entries_with_gpa,

4

COUNT(CASE WHEN gpa > 0 AND gpa <= 2 THEN 1 END) AS suspicious_low_gpa_count,

5

COUNT(CASE WHEN gpa > 4 AND gpa <= 5 THEN 1 END) AS possible_5_point_scale_count,

6

COUNT(CASE WHEN gpa > 5 AND gpa <= 10 THEN 1 END) AS possible_10_point_scale_count,

7

COUNT(CASE WHEN gpa > 10 THEN 1 END) AS invalid_high_gpa_count

8

FROM applicants;

9

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 1

	total_entries bigint	entries_with_gpa bigint	suspicious_low_gpa_count bigint	possible_5_point_scale_count bigint	possible_10_point_scale_count bigint	invalid_high_gpa_count bigint
1	30000	29900	10	30	617	0

After applying a normalized GPA calculation, the average changed from 1.62 to 3.71. This showed me that even a simple metric like GPA cannot be treated as straightforward when the data comes from anonymous, self-submitted entries. Without normalization, the analysis would have produced a technically valid result but a misleading interpretation.

Query		Query History	
1	SELECT		
2	ROUND(AVG(gpa)::numeric, 2) AS raw_average_gpa,		
3	ROUND(AVG(		
4	CASE		
5	WHEN gpa IS NULL THEN NULL		
6	WHEN gpa > 2 AND gpa <= 4 THEN gpa		
7	WHEN gpa > 4 AND gpa <= 5 THEN (gpa / 5.0) * 4.0		
8	WHEN gpa > 5 AND gpa <= 10 THEN (gpa / 10.0) * 4.0		
9	ELSE NULL		
10	END		
11	::numeric, 2) AS normalized_average_gpa		
12	FROM applicants;		
13			
Data Output		Messages	
		Showing rows: 1 to 1	
	raw_average_gpa	normalized_average_gpa	
	numeric	numeric	
1	1.62	3.71	

This differs from official or standardized datasets because official admissions data would usually define the measurement scale, validate the fields, and apply consistent reporting rules. In the Grad Café dataset, the structure is much looser. Applicants can leave fields blank, enter values from different grading systems, or include important context only in the comments. This also helps explain why other metrics, such as GRE Quant, may appear unusually high compared with broader testing averages.